

# RETINXPLAIN

EXPLAINABLE AI FOR DIABETIC RETINOPATHY DETECTION  
AND GRADING USING A USER-CENTRIC APPROACH

*Smridhi Patwari*  
*95-891 Introduction to Artificial Intelligence*



# WHAT IS DIABETIC RETINOPATHY?

## Definition

Microvascular complication of diabetes causing progressive damage to retinal capillaries and macula.

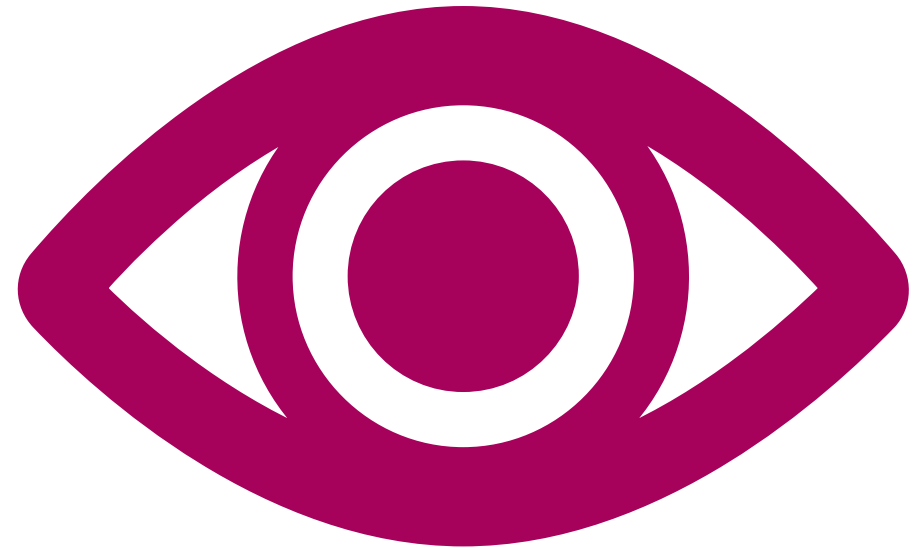
## Who is at risk?

- Long duration of diabetes and insulin-dependent Type 1 diabetes increase risk significantly
- Poor past glycemic control (**HbA1c**)
- Co-morbidities: obesity, hypertension, kidney disease / nephropathy, dyslipidemia, pregnancy

## Early vs Late Presentation

- **Early:** often *asymptomatic* – DR may be found incidentally on routine fundus exam
- **Progressive symptoms** – blurred or distorted vision, floaters, partial vision loss → total vision loss

**Reference:** Shukla UV, Tripathy K. *Diabetic Retinopathy*. StatPearls Publishing; 2025.



# GLOBAL BURDEN OF DIABETIC RETINOPATHY

## Where DR Burden is Highest

### Asia

- *India, China, Southeast Asia*
- Largest diabetes populations worldwide → massive DR prevalence
- Limited access to screening in rural areas

### Sub-Saharan Africa

- Rapidly rising diabetes rates
- Very low screening infrastructure and ophthalmologist shortages

### Latin America

- Higher diabetes prevalence + fragmented healthcare access

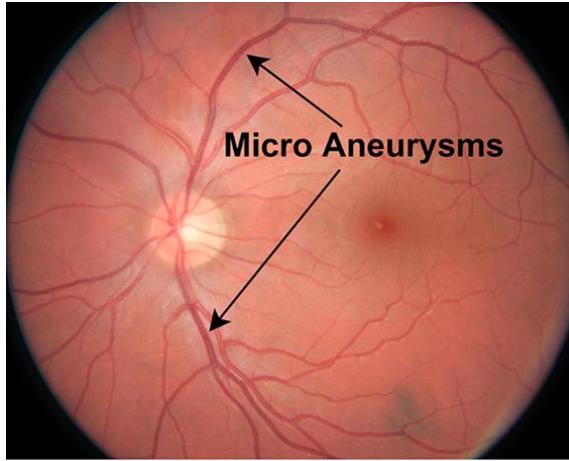
### Pacific Islands & Indigenous Communities

- Extremely high diabetes rates due to genetic & lifestyle factors
- Minimal screening programs

## Why These Regions Are Most Affected?

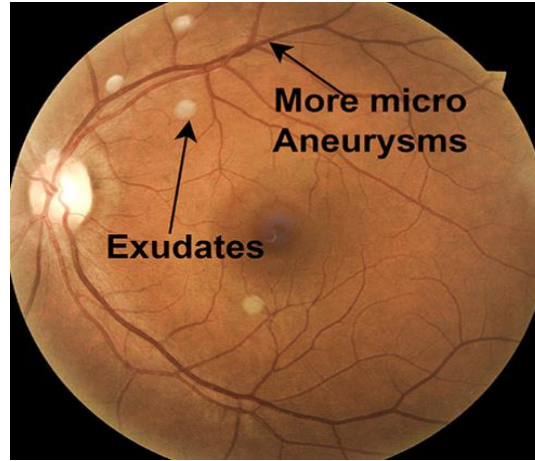
- **Exploding diabetes prevalence** (urbanization, diet changes, aging populations)
- **Low screening coverage** → DR often detected **late**
- **Poor glycemic control** due to cost, medication availability, or health literacy
- **Co-morbidities** like hypertension & obesity increasing disease severity

# THE STAGES OF DR CLASSIFICATION



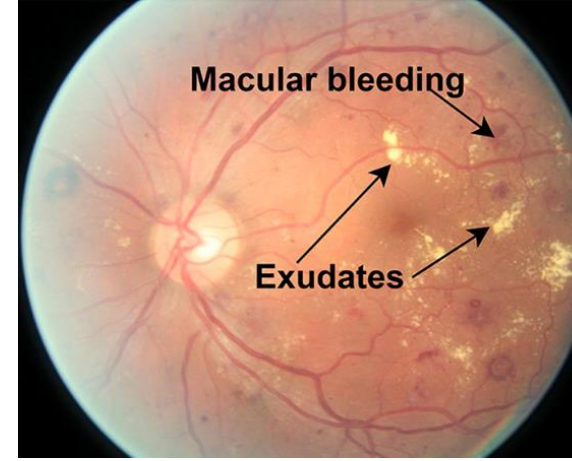
## 1 – Mild DR

- **Microaneurysms only** (tiny red dots)
- No significant hemorrhages
- No cotton-wool spots or IRMA
- Very subtle – easily missed on routine exam



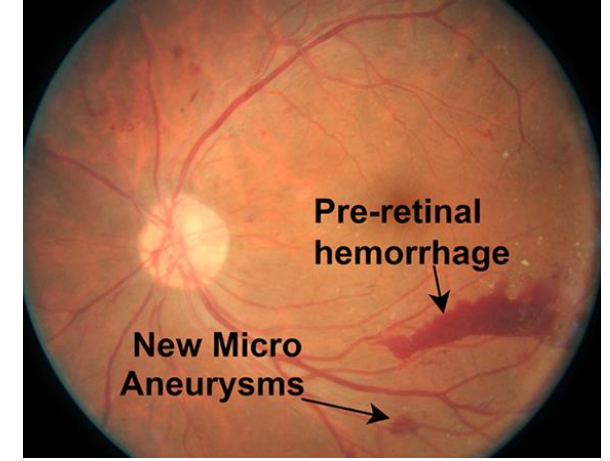
## 2 – Moderate DR

- **More numerous microaneurysms + dot-blot hemorrhages**
- May see **hard exudates**
- May have **cotton-wool spots**
- Lesions visible in **multiple quadrants**, but not extensive



## 3 – Severe DR (Very Advanced Non-Proliferative DR)

- Lots of **retinal bleeding** across many areas
- Blood vessel walls are getting **weaker and irregular**
- New "abnormal" vessels start forming inside the retina (but not yet growing out)
- A warning stage – **very high risk of progressing to PDR**



## 4 – Proliferative DR (PDR)

- **New, fragile blood vessels grow abnormally** on the retina or optic disc
- These vessels can **bleed into the eye** → causing floaters or vision loss
- Most dangerous stage and needs urgent referral/treatment

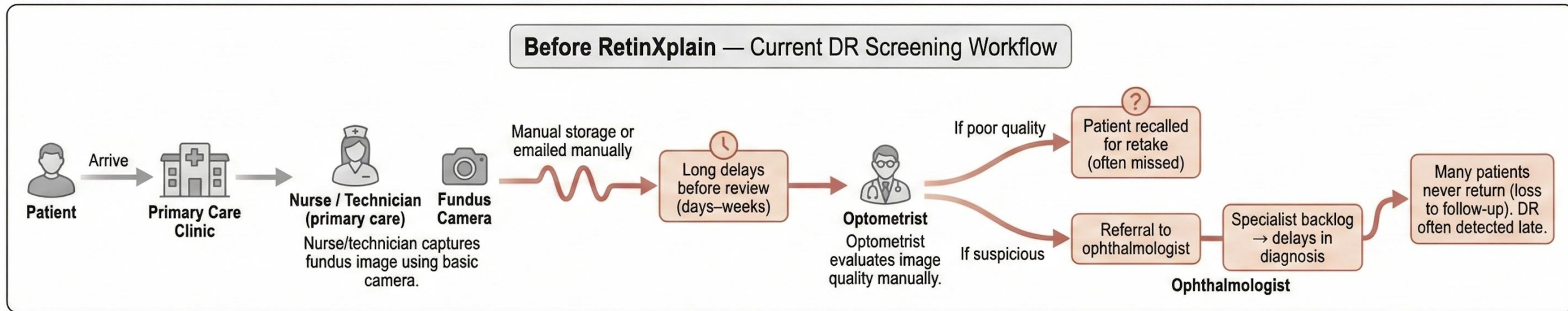
# WHY DR SCREENING IS SLOW & DIFFICULT IN LOW-INCOME COUNTRIES

- High-burden regions (South Asia, Sub-Saharan Africa, Latin America) often lack onsite optometrists or ophthalmologists.
- Patients wait **hours** or travel far for a single retinal image.
- Images are captured by **nurses/technicians** using **basic portable fundus cameras** (smartphone adapters like *Peek Retina*, *Remidio*, or handheld devices like *Volk Pictor Plus*).
- These devices frequently produce **variable-quality images** (small pupils, poor lighting, motion blur).
- Images are often **stored or emailed manually**, causing **long delays** before specialist review.



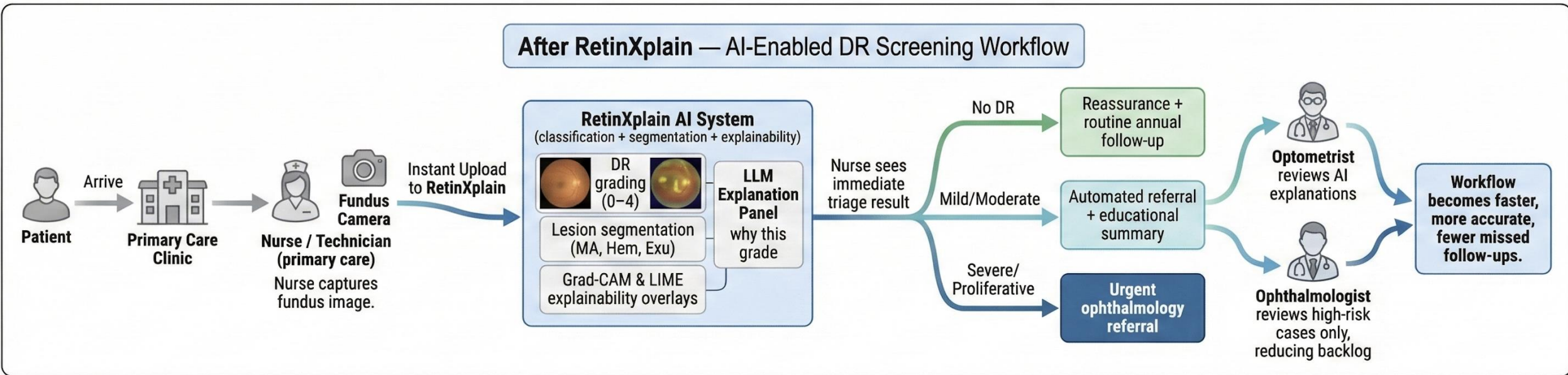


# TRADITIONAL DR SCREENING USER FLOW



**An estimated of 3-12 weeks**  
For a complete diagnosis pathway

# RETINXPLAIN USER FLOW



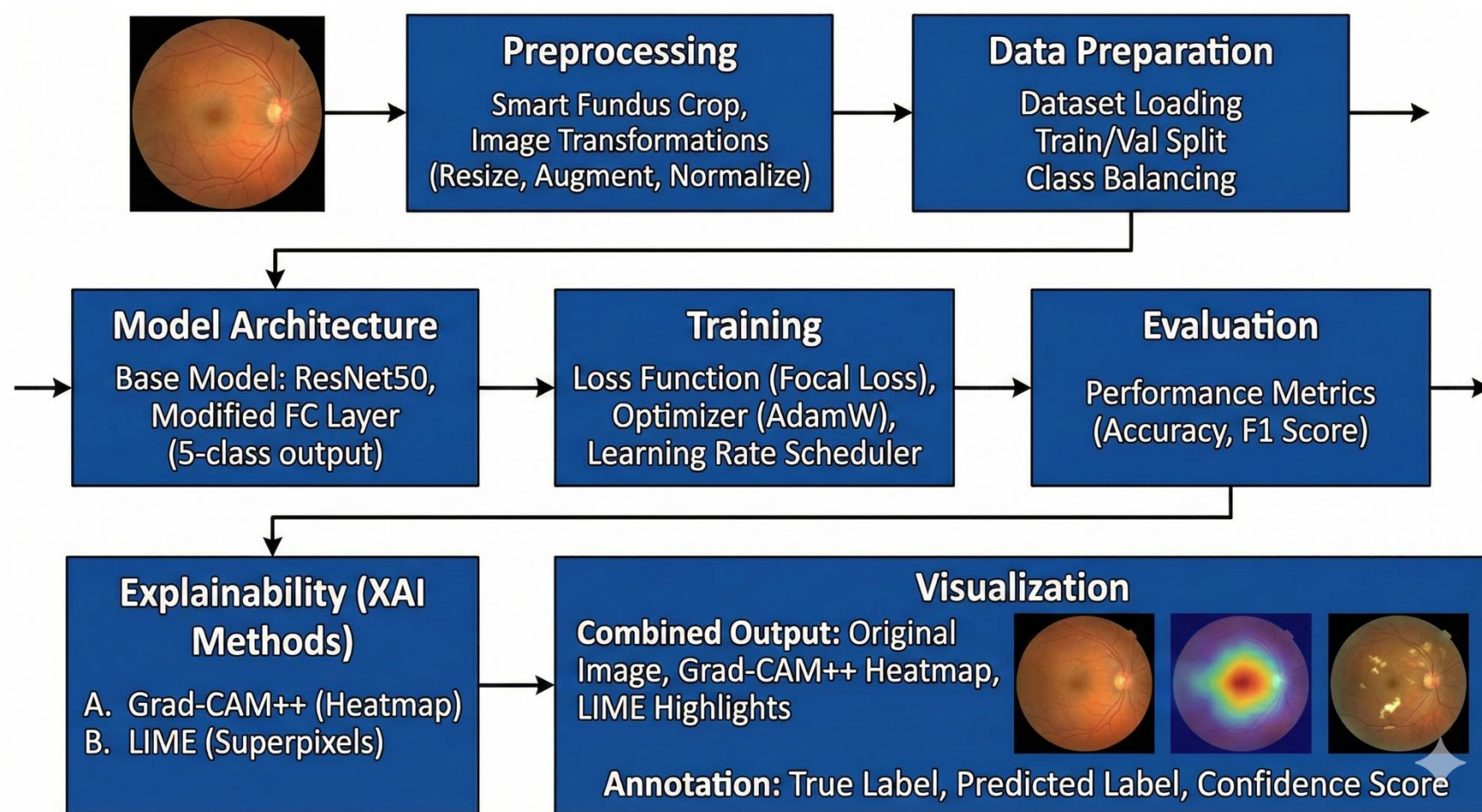
**Same day to 1-3 days, depending on clinic**

For a complete diagnosis pathway

# **MODEL TRAINING, VALIDATION, AND TEST RESULTS**



# MODEL ARCHITECTURE



# CRITERIA FOR SUCCESS



- *Achieve  $\geq 70\%$  validation accuracy.*



- *Grad-CAM heatmaps correspond to medically meaningful regions.*



- *Positive interpretability feedback from mock users.*

# DATASET + DATA SPLIT

## Dataset Split:

- **APTOS 2019 dataset**
- **80/20 stratified split** to preserve the proportion of all DR grades

## How Class Imbalance Was Addressed: **WeightedRandomSampler**

- Ensures minority classes appear more often in each batch
- Prevents the model from being dominated by the majority (No DR)

## Focal Loss ( $\gamma = 2.0$ )

- Down-weights easy examples
- Forces the model to focus on hard/minority classes (Mild, Severe, PDR)

## Augmentations

- Random crop, rotation, flips, color jitter
- Increases variability & reduces overfitting
- Helps minority classes generalize better

| DR CLASS             | COUNT | DESCRIPTION                                      |
|----------------------|-------|--------------------------------------------------|
| 0 - No DR            | 1,434 | Majority class (healthy retina)                  |
| 1 - Mild DR          | 300   | Very subtle lesions; inherently harder to detect |
| 2 - Moderate DR      | 808   | Larger lesion burden; more distinguishable       |
| 3 - Severe DR        | 154   | Fewer samples, high clinical importance          |
| 4 - Proliferative DR | 234   | Rare, advanced disease with neovascularization   |

**Model Accuracy: 79.2%**

**Overall Weighted F1: 0.781**

**Recall: 0.781**

# EXPLAINABILITY

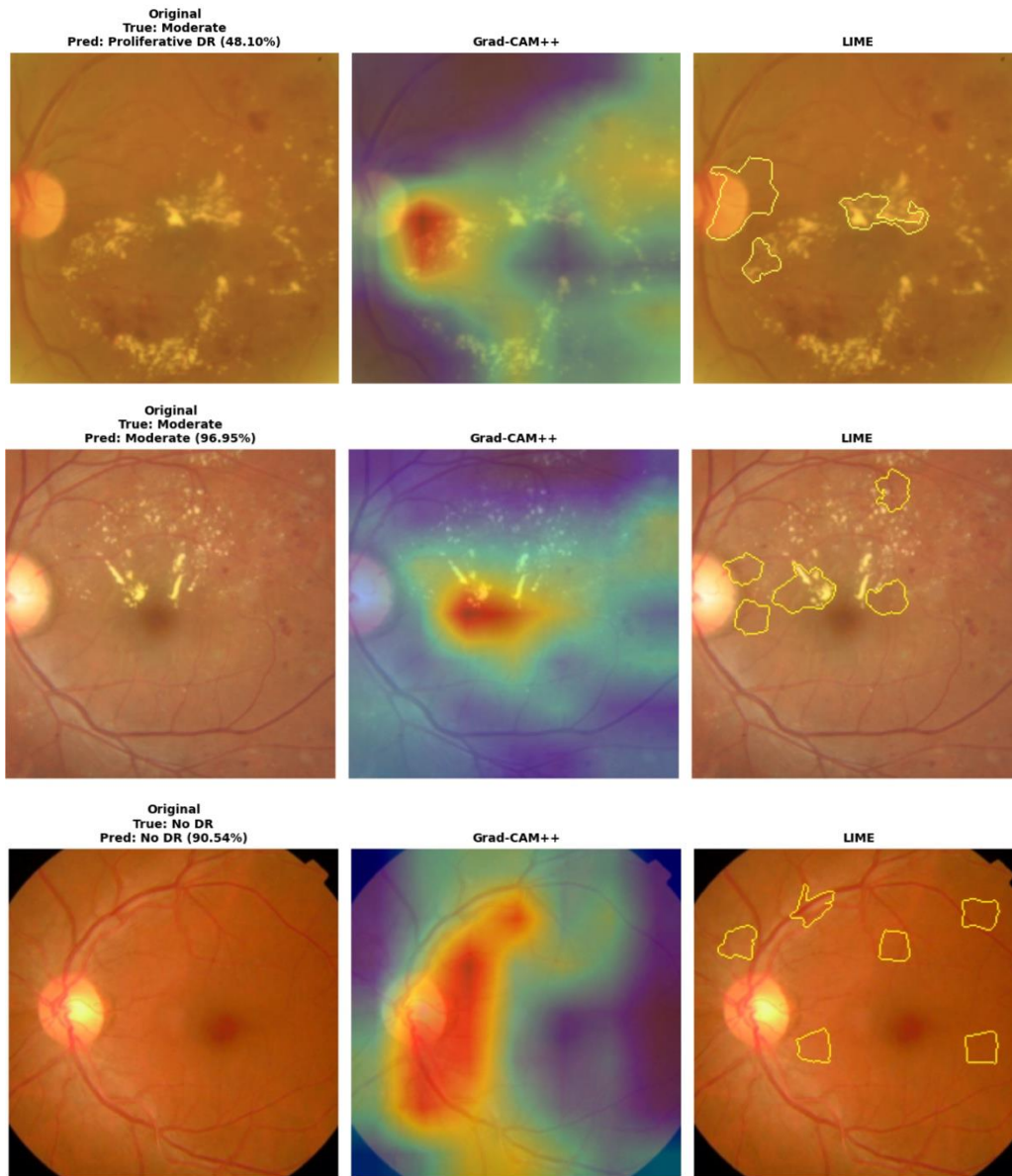
## Grad-CAM++

- Accurately highlights **pathological regions** (e.g., hard exudates, neovascular tufts, cotton-wool spots)
- Focus maps align well with real DR lesions
- Shows that the model is learning **clinically meaningful features**

## LIME

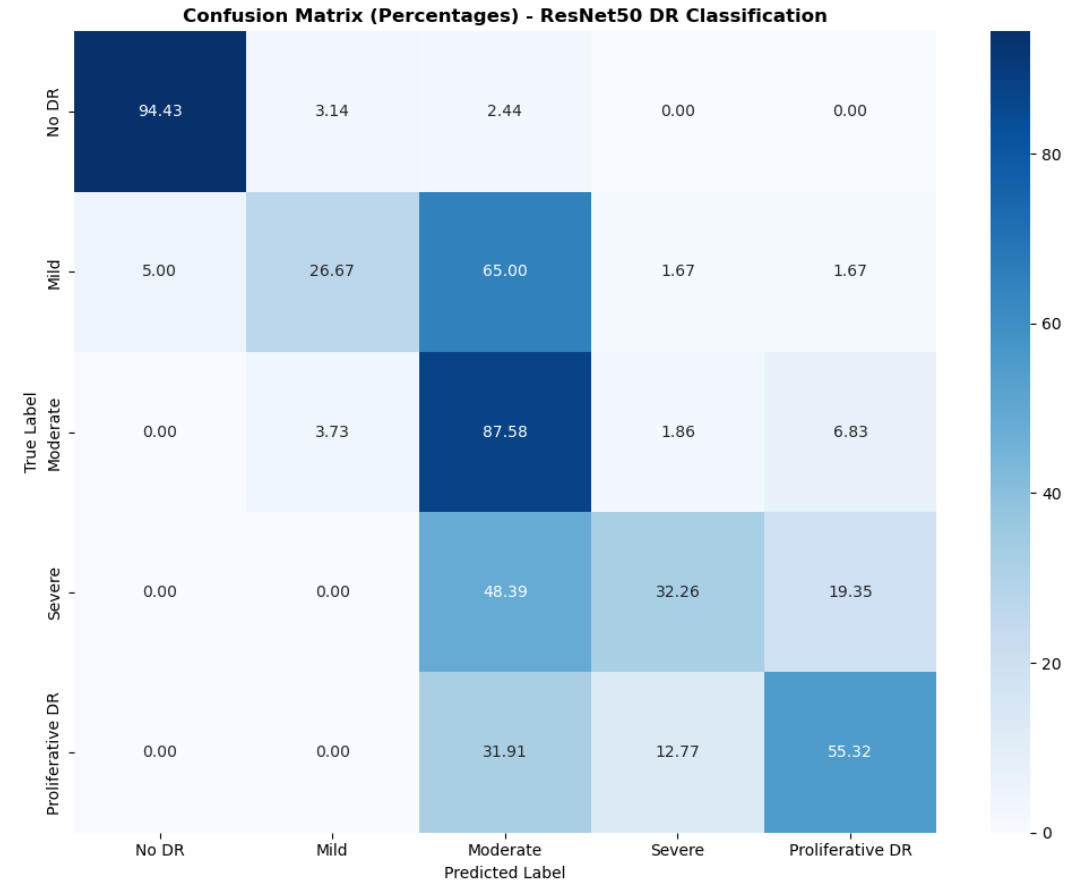
- Produces segmented superpixel regions around **distinct lesion clusters**
- Clear for Moderate-Proliferative DR where lesions have shape/texture contrast
- Supports lesion-level interpretability (useful for referral justification)

*Explainability is most valuable for diseased eyes. In healthy images, heatmaps show attention—not pathology.*



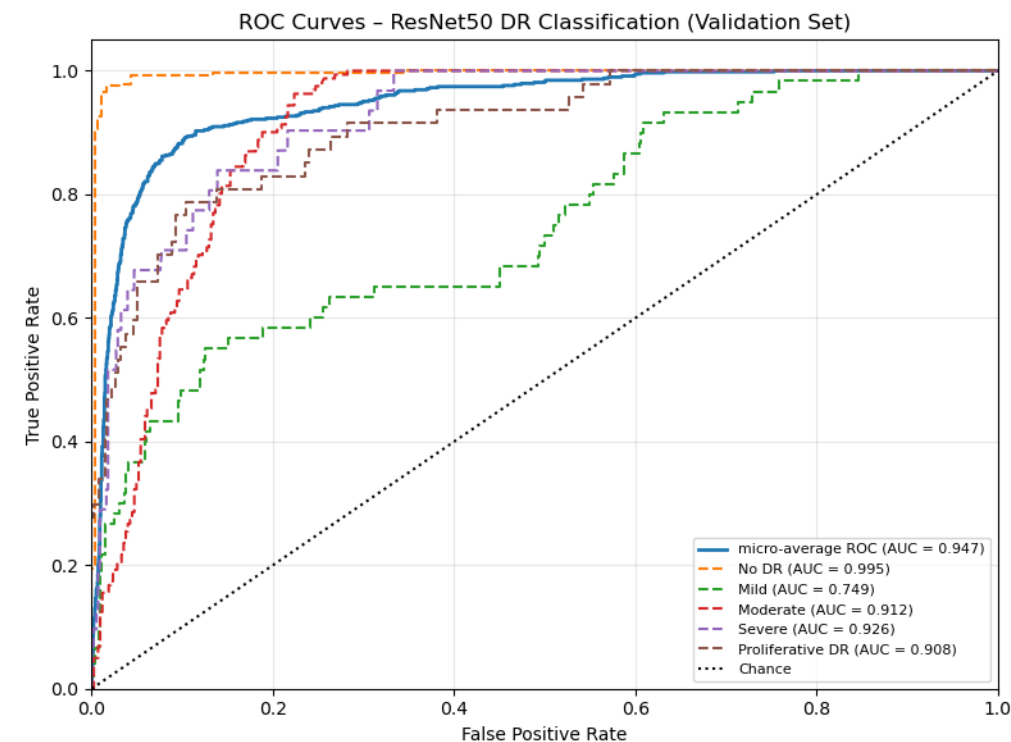
# CONFUSION MATRIX

- The model shows **strong overall performance**, with most predictions aligned along the diagonal.
- **Best-performing class: No DR**, likely due to a large number of samples in this category.
- **Most errors occur in the Mild and Severe classes**, which are often **misclassified as Moderate DR**—reflecting subtle boundaries between these stages and limited samples.





# ROC CURVES



| Metric       | Value  | Interpretation                                                                                                                   |
|--------------|--------|----------------------------------------------------------------------------------------------------------------------------------|
| Macro AUC    | 0.8978 | Average performance across all classes, treating each class equally. Strong overall model performance even with class imbalance. |
| Micro AUC    | 0.9471 | Overall discriminative ability across all samples. Indicates the model ranks predictions very effectively.                       |
| Weighted AUC | 0.9360 | High score shows model performs well even with imbalanced classes (e.g., Mild/Severe being less common).                         |

| DR Class         | AUC Score | Interpretation                                                                                                                                |
|------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| No DR            | 0.9945    | Outstanding discrimination. Model almost perfectly separates healthy eyes from any DR stage. Ideal for screening.                             |
| Mild DR          | 0.7489    | Fair discrimination. Mild lesions are subtle; overlapping features with No DR and Moderate cause lower separability. Expected in DR datasets. |
| Moderate DR      | 0.9115    | Excellent discrimination. Model reliably identifies moderate DR due to more obvious hemorrhages and microaneurysm clusters.                   |
| Severe DR        | 0.9263    | Excellent performance. Strong ability to detect severe disease, which is clinically important for urgent referral.                            |
| Proliferative DR | 0.9076    | Very good discrimination. Despite fewer samples, model detects PDR features like neovascularization confidently.                              |

**APP DEMO**

# KEY TAKEAWAYS

- **Class imbalance heavily shapes model behavior.**  
Without balanced training, the model learns to over-rely on majority classes, giving artificially high accuracy while still missing true disease cases.
- **Explainability exposed what accuracy alone could not.**  
Before applying any class imbalance handling, Grad-CAM revealed that the model was not attending to clinically relevant regions, even with high accuracy scores. This highlights why model explainability is essential in medical AI.
- **Balancing the dataset changed both where the model looked and how it predicted.**  
After introducing class weighting / sampling strategies, the model's attention maps aligned more closely with real DR lesions (microaneurysms, hemorrhages, exudates), demonstrating better clinical grounding.

# FUTURE WORK

- **Further fine-tuning and model comparison**  
Evaluate ResNet-50, EfficientNet-B0, and EfficientNet-B4 to identify the best balance between
- **Select a lightweight model for real-world deployment**  
Since RetinXplain is designed for low-resource and low-connectivity settings, the chosen model must run efficiently **on-device** with minimal computational load.
- **Extend to a unified multi-disease classifier**  
Move from single-disease DR detection to a multi-label model capable of identifying **DR, Glaucoma, Cataract, and Age-related Macular Degeneration** in one pass.
- **Improve robustness with realistic augmentation**  
Incorporate motion blur, Gaussian blur, uneven lighting, and image compression artifacts to simulate low-cost fundus cameras and real field conditions.
- **Integrate explicit Image Quality Assessment (IQA)**  
Add an automated quality grading module in the UI to flag ungradable images and prompt retakes before DR prediction.

**THANK  
YOU**

